

Weiwu Yan

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Education

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| University of Chicago | Fall 2023 – Spring 2025 |
| <ul style="list-style-type: none">• Master of Arts in Computational Social Science | |
| University of California, San Diego | Fall 2019 – Spring 2023 |
| <ul style="list-style-type: none">• Bachelor of Science in Social Psychology• Bachelor of Arts in Sociology | |

Projects

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| Gender & Video Game Dialogue (in progress) | Spring 2025 |
| <ul style="list-style-type: none">• Collected and processed 247,644 dialogue lines from 3,043 characters across 18 role-playing video games using the Video Game Dialogue Corpus.• Conducted emotion classification (RoBERTa) to analyze gender differences in emotional expression.• Performed Part-of-Speech (spaCy) tagging and TF-IDF vectorization (scikit-learn) to identify the verbs most uniquely associated with each gender.• Utilized regular expressions to examine gender differences in the distribution of terminal punctuation. | |

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| A Chat Application | Fall 2024 |
| <ul style="list-style-type: none">• Designed the frontend of a chat application (HTML & CSS) with features such as a navigation bar, interactive hover effects, and responsive layout adaptation.• Developed the backend of a chat application (Flask) with features such as user account management, channel-based communication, and real-time messaging for text, emojis, and image sharing. | |

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| Scholarly Article Classification | Spring 2024 |
| <ul style="list-style-type: none">• Collected and processed metadata of 1.7 million scholarly articles from the arXiv Dataset.• Performed Multinomial Logistic Regression (Apache Spark) on AWS to identify latent topics across articles.• Trained a Latent Dirichlet Allocation model (Apache Spark) on AWS to classify articles into research fields. | |

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| Urban Gentrification | Spring 2024 |
| <ul style="list-style-type: none">• Designed an agent-based model (Mesa) to simulate urban gentrification dynamics, examining how citizens and immigrants with varying socioeconomic status interact and relocate within a city. | |

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| Funding & Academic Output | Winter 2023 |
| <ul style="list-style-type: none">• Collected and processed 4,959 funding records from U.S. NSF Division of Behavioral and Cognitive Sciences.• Scraped metadata of 207,370 scholarly articles authored by 3,230 identified researchers from Google Scholar.• Applied k-means clustering and TF-IDF vectorization (scikit-learn) to examine the relationship between seven identified fields and funding amounts.• Utilized a Structural Topic Model (R) to analyze temporal trends across 12 identified topics.• Performed document embeddings (SciBERT) to assess the impact of funding on research diversity. | |

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| Video Game Classification | Winter 2023 |
| <ul style="list-style-type: none">• Collected and processed metadata of 75,776 video games from the Steam Games Dataset.• Trained a Random Forest Classifier (scikit-learn) to classify games based on textual descriptions.• Trained a Convolutional Neural Network (TensorFlow) to classify games based on cover art. | |

Eyewitness Identification

Summer 2022

- Collected and processed responses from 78 participants in a between-subjects factorial design experiment (Qualtrics).
- Conducted an ANOVA test (Jamovi) to analyze the effects of lineup procedure, target availability, and confidence on eyewitness identification accuracy.

Moral Judgment & Occupation

Fall 2020

- Collected and processed responses from 285 participants in an observational study (Qualtrics).
- Conducted an ANOVA test (Jamovi) to analyze the relationship between maternal occupation and moral judgments of mothers leaving their children unsupervised.

Skills

Languages: Chinese & English

Programming: Python, SQL, JavaScript & R

Artificial Intelligence: TensorFlow, PyTorch, spaCy, scikit-learn, Natural Language Toolkit & Keras

Data Analysis: Pandas, NumPy, SciPy & dplyr

Data Visualization: Plotly, ggplot2, Tableau & Matplotlib

High-Performance Computing: Amazon Web Services & Apache Spark

Web Development: Shiny, Flask & React

Survey: Qualtrics & Google Survey